Yewon Han

PhD Student | Trustworthy AI and AI Safety

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OnedayOneyeah

Research Profile

- PhD student at the Learning and Adaptation Algorithms Lab (LAAL), Seoul National University, advised by Prof. Taesup Kim. Research focuses on trustworthy AI, uncertainty estimation, robust adaptation, and safety in vision-language and egocentric multimodal assistants.

Education

Seoul National University — Seoul, South Korea
Integrated MS/PhD in Data Science — 2023 – present
- Entered the integrated MS/PhD program in 2023; began the doctoral stage in 2025.
- Learning and Adaptation Algorithms Lab (LAAL); Advisor: Prof. Taesup Kim.

Seoul National University — Seoul, South Korea
Bachelor's degree in Communication — 2018 – 2023

University of Toronto — Toronto, Canada
Exchange Student in Faculty of Information — 2021 – 2022

Research Interests

- AI safety, adversarial robustness, and safety–utility trade-offs in large vision-language models.
- Uncertainty estimation, energy-based models, and test-time adaptation.
- Safety evaluation of wearable and egocentric multimodal assistants; efficient on-device AI.

Awards

Research Support Program in Science and Engineering (Master's Student Research Scholarship), NRF — Sept 2024

Best Paper Award (Artificial Intelligence) — June 2024
"Test-time Adaptation with Energy-based Indicator in Open-set Scenarios," Korea Computer Congress (KCC) 2024.

Google ExploreCSR Program — Second Prize — Feb 2023
Developed a KoBERT-based Korean fact verification system in the Google Research-sponsored ExploreCSR Program, earning second prize. This experience sparked my initial interest in AI safety and reliable human–AI interaction.

FactCheck Cover Contest — Google News Prize — Oct 2020
Team fact-checking project on medical school enrollment, doctors' income, and regional data visualization.

Investigative Journalism Cover Contest — Second Prize — Oct 2019
Team investigative reporting on older adults collecting waste paper and gaps in social protection.

Publications

Safety for Whom? Benchmarking Understanding Beyond Refusal in Egocentric Multimodal Assistants — Oct 2026
Yewon Han, Jisu Jeon, Suhyun Choi, Juan Yeo, Minsoo Jo, Gihoon Kim, Hyunggyu Lim, Taesup Kim
arxiv.org/abs/2610.17728 (arXiv preprint, arXiv:2610.17728)

Two Birds, One Projection: Harmonizing Safety and Utility in LVLMs via Inference-Time Feature Projection — 2026
Yewon Han, Yumin Seol, Eun Gyung Kong, Minsoo Jo, Taesup Kim
doi.org/10.1007/978-3-032-37167-6_18 (European Conference on Computer Vision (ECCV), 2026)

Contrastive Residual Energy Test-time Adaptation — May 2025
Revised May 2026.
Yewon Han, Seoyun Yang, Taesup Kim
arxiv.org/abs/2505.19607 (arXiv preprint, arXiv:2505.19607)

Test-time Adaptation with Energy-based Indicator in Open-set Scenarios — June 2024
Best Paper Award (Artificial Intelligence).
Yewon Han, Kyubyung Chae, Taesup Kim
www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11861935 (Proceedings of the Korean Information Science Society (KCC 2024))

Projects

Hanwha Systems — Oct 2026 – present
Industry–academia collaborative research projects.

Qualcomm University Relations — May 2026 – present
Training-aware emulation for efficient on-device training.

Development of Fundamental AI Technologies Inspired by Human Cognition — Mar 2026 – present
FPN-inspired metacognitive module for uncertainty-driven dynamic adjustment of reasoning paths and computational budgets.

factcheck-ko-2023 — Jan 2023 – Jan 2024
Korean fact-checking as a natural language inference (NLI) information verification task at Seoul National University.

Professional Experience

Marketing Intern: NAVER WEBTOON

Teaching

Computing Fundamentals: Teaching Assistant — Spring 2024, Fall 2024, Fall 2025, Spring 2026

Computing Fundamentals for Data Science: Teaching Assistant — Spring 2025

Deep Learning II for Data Science: Teaching Assistant — Spring 2026

Languages

Korean
Native proficiency

English
Advanced proficiency